

SAM-V2: Agentic Model Concepts (Interview Guide)

This document outlines the theoretical shift from a Generative Chatbot to an Autonomous Agent. These concepts form the backbone of modern AI automation.

1. Function/Tool Calling

The fundamental capability of an Agent. We provide the LLM with a schema of available Python tools. The LLM processes the user query and autonomously decides precisely which function to execute and what arguments to pass, entirely removing the need for 'router.py'.

2. Structured Output Binding (JSON)

Standard LLMs output conversational text. An Agent must output deterministic, machine-readable data. We enforce the model to return strict JSON strings that the backend can parse directly into programmatic actions.

3. The ReAct Framework (Reason, Act, Observe)

Agents operate in loops. They Reason ('the user wants to scan the network'), they Act (trigger the NMAP tool), and then they Observe the terminal output. This internal dialogue loop allows the Agent to correct its own mistakes and handle multi-step chains.

4. Stateful Context Memory

Generative models are inherently stateless (they forget after one interaction). An Agentic model implements a rolling memory buffer (message arrays) appended to every query, enabling deep conversational context and sustained reasoning.